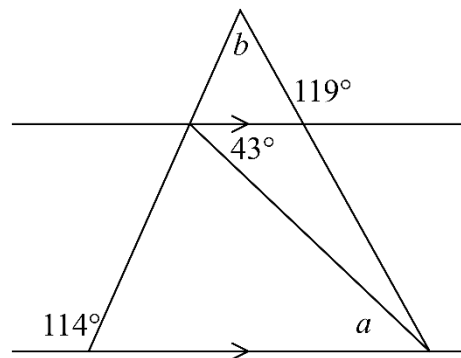


Section I (10 marks)

- 1 Which of the following is equivalent to $\sqrt{243} + 2\sqrt{75}$?
- (A) $19\sqrt{3}$
- (B) $81\sqrt{3}$
- (C) $106\sqrt{3}$
- (D) $2\sqrt{318}$
- 2 What is the domain of the function $f(x) = \frac{1}{(x-3)(1-x)}$?
- (A) $x \neq -1$ or $x \neq -3$
- (B) $x \neq -1$ or $x \neq 3$
- (C) $x \neq 1$ or $x \neq -3$
- (D) $x \neq 1$ or $x \neq 3$
- 3 What is the value of $\frac{dy}{dx}$ if $y = x^4 + 5x^{-1}$?
- (A) $\frac{dy}{dx} = 4x^3 - 5x^0$
- (B) $\frac{dy}{dx} = 4x^3 + 5x^0$
- (C) $\frac{dy}{dx} = 4x^3 - 5x^{-2}$
- (D) $\frac{dy}{dx} = 4x^3 + 5x^{-2}$
- 4 Solve $|2 - 3x| \geq 5$
- (A) $x \leq -1$ or $x \geq 2\frac{1}{3}$
- (B) $x \geq -1$ or $x \leq 2\frac{1}{3}$
- (C) $x \leq -2\frac{1}{3}$ or $x \geq 1$
- (D) $x \geq -2\frac{1}{3}$ or $x \leq 1$

- 5 Which type of function is $f(x) = 2x^3 - x$?
- (A) Odd
(B) Even
(C) Neither odd or even
(D) Zero
- 6 What is solution to the equation $2\sin \beta = -\sqrt{3}$ for $0^\circ \leq \beta \leq 360^\circ$?
- (A) $\beta = 60^\circ$ or 300°
(B) $\beta = 120^\circ$ or 240°
(C) $\beta = 210^\circ$ or 330°
(D) $\beta = 240^\circ$ or 300°
- 7 What is the solution to the equation $x^2 + 2x - 7 = 0$?
- (A) $x = -1 \pm \sqrt{2}$
(B) $x = -2 \pm \sqrt{2}$
(C) $x = -2 \pm 2\sqrt{2}$
(D) $x = -1 \pm 2\sqrt{2}$
- 8 What are the values of a and b ?



Not to scale

- (A) $a = 43^\circ$ and $b = 53^\circ$
(B) $a = 43^\circ$ and $b = 61^\circ$
(C) $a = 45^\circ$ and $b = 53^\circ$
(D) $a = 45^\circ$ and $b = 61^\circ$

9 The line $4x - ky = 3$ passes through the point $(1, 2)$. What is the value of k ?

(A) -2

(B) $-\frac{1}{2}$

(C) $\frac{1}{2}$

(D) 2

10 What is $pqx^2 - p^3q$ fully factorised?

(A) $pq(x + p)(x - p)$

(B) $p^2(x + q)(x - q)$

(C) $pq(x^2 - p^2)$

(D) $p^2(x^2 - q^2)$

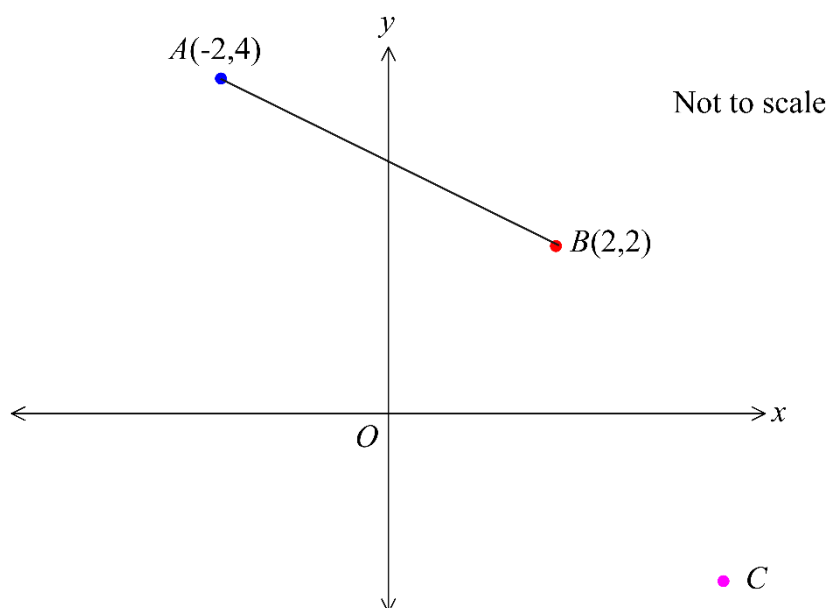
Section II (60 marks)

Question 11 (15 marks)

Use a SEPERATE sheet of paper

Marks

- (a) Express $0.1\dot{5}$ as a fraction in its simplest form. 2
- (b) Solve $\frac{1}{2}(y-3) - \frac{1}{3}(y-2) = 3$ 2
- (c) The cost of a smart watch is \$979. This includes a 10% tax on the original price. Calculate the original price of the smart watch. 1
- (d) Simplify $\frac{a^2 + b^2 + 2ab}{-b^2 + a^2}$ 2
- (e) Find the exact value of:
- (i) $\tan 240^\circ$ 1
- (ii) $\sec 510^\circ$ 1
- (f) The diagram shows the points $A(-2,4)$, $B(2,2)$ and $O(0,0)$.
Point C is the fourth vertex of the parallelogram $OABC$.



- (i) Find the equation of the straight line that passes through the points A and B. 2
- (ii) Show that the length of AB is $2\sqrt{5}$. 1
- (iii) Calculate the perpendicular distance from O to the line AB . 1
- (iv) Calculate the area of parallelogram $OABC$. 1
- (v) What are the coordinates of C ? 1

Question 12 (13 marks)

Use a SEPERATE sheet of paper

Marks

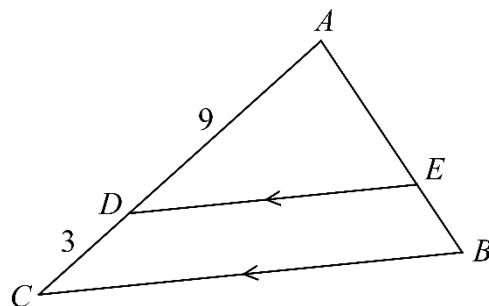
(a) Simplify $\frac{x}{x^2-4} + \frac{3}{x-2}$

2

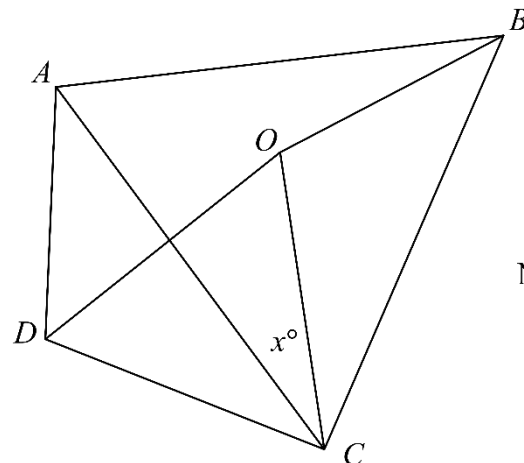
(b) Rationalise the denominator of $\frac{1}{\sqrt{5}-1}$

2

(c) Expand and then simplify $(1-y)^3 + (1+y)^2$

2(d) ABC is a triangle with DE drawn parallel to BC .

Not to scale

Given $AD=9$, $DC=3$ and $AB=8$.(i) Show that $\triangle ABC \parallel \triangle ADE$.**2**(ii) Find the length of BE .**2**(e) In the diagram, $\triangle ABC$ and $\triangle ODC$ are equilateral and $\angle OCA = x^\circ$.

Not to scale

(i) Show that $\angle ACD = \angle BCO$ **1**(ii) Prove that $\triangle ACD \equiv \triangle BCO$ **2**

Question 13 (16 marks)

Use a SEPERATE sheet of paper

Marks

(a) Solve the following equations.

(i) $x^2 - 16 = 0$ 1

(ii) $25^k \cdot (5^3)^4 = 1$ 2

(b) Factorise completely

(i) $y^3 + 10y^2 + 25y$ 1

(ii) $a^3 - \frac{1}{8}$ 1

(c) Draw neat sketches of the following equations on a separate set of axes.
Show clearly the essential features of each graph.

(i) $y = 4 - x^2$ 1

(ii) $y = \sqrt{4 - x^2}$ 1

(iii) $y = \frac{4}{x}$ 1

(d) Find $\lim_{x \rightarrow 4} \frac{x^2 - x - 12}{x - 4}$ 2

(e) The function $y = f(x)$ is defined as follows:

$$f(x) = \begin{cases} 3^x & \text{for } x > 0 \\ 1 & \text{for } x = 0 \\ x^{-1} & \text{for } x < 0 \end{cases}$$

Find the value of $f(3) + f(0) + f(-3)$. 2

(f) The curve $y^2 = x + 9$ and the straight line $x - 3y + 9 = 0$ intersect at A and B .

(i) Find the coordinates of A and B . 2

(ii) Find the values of x for which $24 + 2x - x^2 \leq 0$. 2

Question 14 (16 marks) Use a SEPERATE sheet of paper **Marks**

(a) Prove $\tan \theta - \tan \theta \sin^2 \theta = \sin \theta \cos \theta$ **2**

(b) Solve the following pair of simultaneous equations. **2**

$$2x + y - 1 = 0$$

$$3x - y - 4 = 0$$

(c) Differentiate with respect to x .

(i) $\sqrt{x^4 + 7}$ **1**

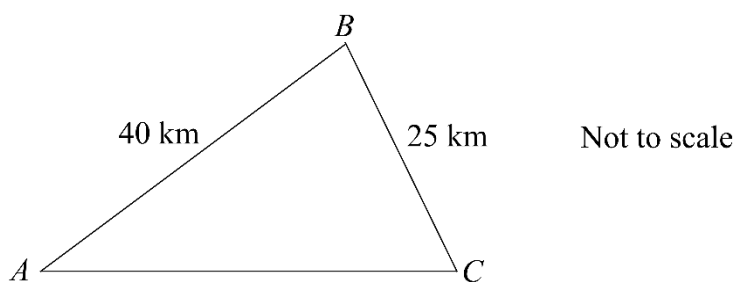
(ii) $x(1 - x)^6$ **2**

(iii) $\frac{3x}{2 - x}$ **2**

(d) Find the equation of the tangent to the curve $y = x^2 - 6x$ at $(-1, 7)$. **2**

(e) Solve the equation $2^x \times 4^{x+1} = \frac{1}{2}$ **1**

(f) Point C is due east of A . Point B is 40 km from A and 25 km from C .
The bearing of B from C is 325° .



(i) What is the size of $\angle ACB$? **1**

(ii) What is the bearing of B from A ? **3**

End of paper

Mathematics AP2 Examination 2016

NAME.....

CLASS 11MT.....

Use the multiple-choice answer sheet.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

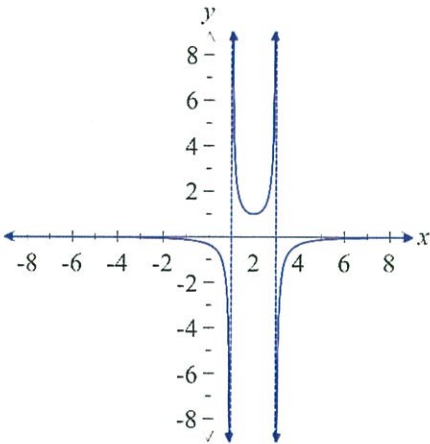
A ☒ B ☒ C ☐ D ☐

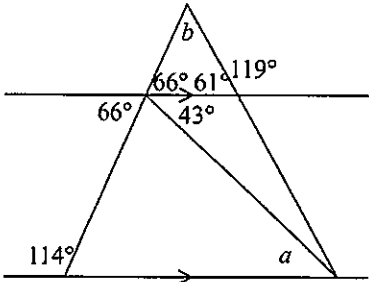
If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
 correct
 ↖

1. A ☐ B ☐ C ☐ D ☐
2. A ☐ B ☐ C ☐ D ☐
3. A ☐ B ☐ C ☐ D ☐
4. A ☐ B ☐ C ☐ D ☐
5. A ☐ B ☐ C ☐ D ☐
6. A ☐ B ☐ C ☐ D ☐
7. A ☐ B ☐ C ☐ D ☐
8. A ☐ B ☐ C ☐ D ☐
9. A ☐ B ☐ C ☐ D ☐
10. A ☐ B ☐ C ☐ D ☐

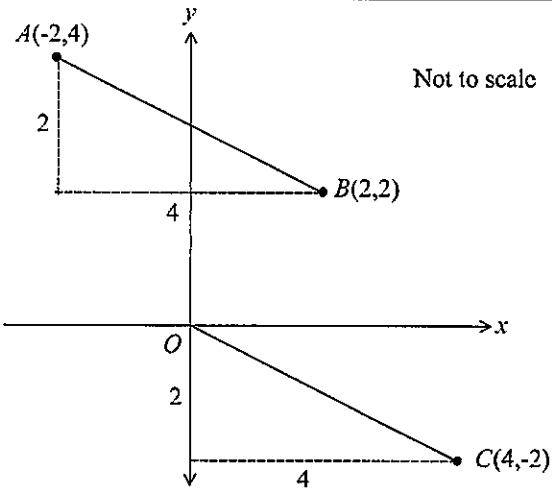
CTHS AP2 Examination 2016
Year 11 Mathematics Yearly Examination
Worked solutions and marking guidelines

Section I		
	Solution	Criteria
1	$\sqrt{243} + 2\sqrt{75} = \sqrt{81} \times \sqrt{3} + 2 \times \sqrt{25} \times \sqrt{3}$ $= 9\sqrt{3} + 10\sqrt{3} = 19\sqrt{3}$	1 Mark: A
2	Domain: $x \neq 1$ or $x \neq 3$ 	1 Mark: D
3	$y = x^4 + 5x^{-1}$ $\frac{dy}{dx} = 4x^3 - 5x^{-2}$	1 Mark: C
4	$2 - 3x \geq 5 \quad \text{or} \quad 2 - 3x \leq -5$ $-3x \geq 3 \quad \quad \quad -3x \leq -7$ $x \leq -1 \quad \quad \quad x \geq \frac{7}{3} \text{ or } 2\frac{1}{3}$ Solution is $x \leq -1$ or $x \geq 2\frac{1}{3}$	1 Mark: A
5	$f(-x) = 2(-x)^3 - (-x)$ $= -(2x^3 - x)$ $= -f(x)$ Therefore, the function is odd.	1 Mark: A
6	$2 \sin \beta = -\sqrt{3}$ $\sin \beta = -\frac{\sqrt{3}}{2}$ $\beta = 240^\circ \text{ or } 300^\circ$	1 Mark: D

7	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ $= \frac{-2 \pm \sqrt{2^2 - 4 \times 1 \times -7}}{2 \times 1}$ $= \frac{-2 \pm \sqrt{32}}{2}$ $= -1 \pm 2\sqrt{2}$	1 Mark: D
8	 $a = 43^\circ$ (alternate angles are equal, two parallel lines) $b + 66 + 61 = 180$ (angle sum of a triangle is 180) $b = 53^\circ$ $a = 43^\circ$ and $b = 53^\circ$	1 Mark: A
9	(1, 2) satisfies the equation $4x - ky = 3$ $4 \times 1 - k \times 2 = 3$ $-2k = -1$ $k = \frac{1}{2}$	1 Mark: C
10	$pqx^2 - p^3q = pq(x^2 - p^2)$ $= pq(x + p)(x - q)$	1 Mark: A

Section II, Question 11 (15 marks)

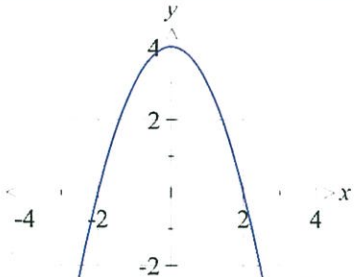
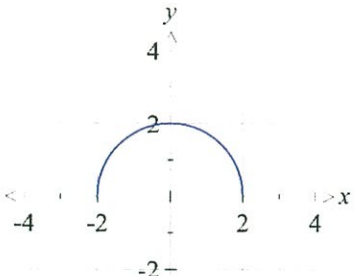
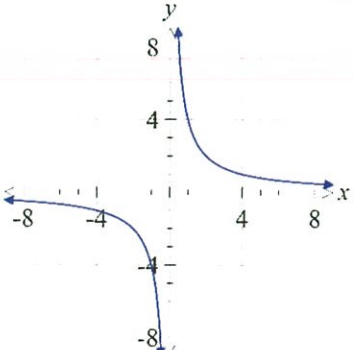
11(a)	Let $x = 0.1\dot{5}$ $x = 0.155555... \quad (1)$ $10x = 1.555555... \quad (2)$ Equation (2) - (1) $9x = 1.4$ $x = \frac{1.4}{9} = \frac{7}{45}$	2 Marks: Correct answer. 1 Mark: Makes some progress towards the solution.
11(b)	$6 \times \left[\frac{1}{2}(y-3) \right] - \left[\frac{1}{3}(y-2) \right] \times 6 = 3 \times 6$ $3(y-3) - 2(y-2) = 18$ $3y - 9 - 2y + 4 = 18$ $y = 23$	2 Marks: Correct answer. 1 Mark: Multiplies both sides by 6 or makes some progress.
11(c)	Let \$x be the original cost. $x + 10\% \text{ of } x = \979 $x + 0.1x = \$979$ $1.1x = \$979$ $x = \$890$	1 Mark: Correct answer.
11(d)	$\frac{a^2 + b^2 + 2ab}{-b^2 + a^2} = \frac{(a+b)^2}{a^2 - b^2}$ $= \frac{(a+b)^2}{(a+b)(a-b)} = \frac{(a+b)}{(a-b)}$	2 Marks: Correct answer. 1 Mark: Factorises one term.
11(e) (i)	$\tan 240^\circ = \tan(180^\circ + 60^\circ)$ $= \tan 60^\circ = \sqrt{3}$	1 Mark: Correct answer.
11(e) (ii)	$\sec 510^\circ = \sec(360^\circ + 150^\circ) = \sec(150^\circ)$ $= \sec(180^\circ - 30^\circ)$ $= -\sec(30^\circ)$ $= -\frac{2}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{-2\sqrt{3}}{3}$	1 Mark: Correct answer.
11(f) (i)	$\text{Gradient } AB = \frac{y_2 - y_1}{x_2 - x_1} = \frac{2 - 4}{2 - (-2)} = \frac{-2}{4} = -\frac{1}{2}$	2 Marks: Correct answer. 1 Mark: Finds the gradient of AB or shows some understanding of the problem.

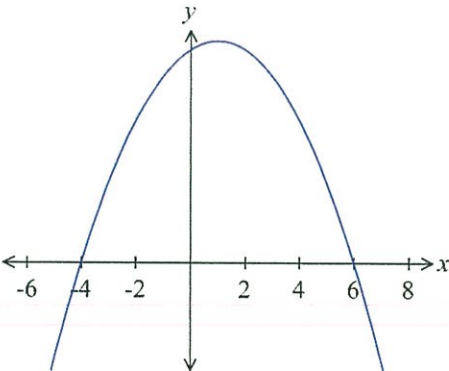
	Equation of AB $y - y_1 = m(x - x_1)$ $y - 2 = -\frac{1}{2}(x - 2)$ $-2y + 4 = x - 2$ $x + 2y - 6 = 0$	
11(f) (ii)	Distance $AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ $= \sqrt{(2 - (-2))^2 + (2 - 4)^2}$ $= \sqrt{20} = 2\sqrt{5}$	1 Mark: Correct answer.
11(f) (iii)	$d = \frac{ ax_1 + by_1 + c }{\sqrt{a^2 + b^2}}$ $= \frac{ 1 \times 0 + 2 \times 0 - 6 }{\sqrt{1^2 + 2^2}}$ $= \frac{ -6 }{\sqrt{5}} = \frac{6}{\sqrt{5}} \text{ or } \frac{6\sqrt{5}}{5}$	1 Mark: Correct answer.
11(f) (iv)	$A = bh = 2\sqrt{5} \times \frac{6}{\sqrt{5}} = 12$ square units	1 Mark: Correct answer.
11(f) (v)	OC and AB are parallel and have the same gradient. $C(4, -2)$  Not to scale	1 Mark: Correct answer.

Question 12 (13 marks)

12(a)	$\frac{x}{x^2-4} + \frac{3}{x-2} = \frac{x}{(x-2)(x+2)} + \frac{3}{(x-2)}$ $= \frac{x+3(x+2)}{(x-2)(x+2)}$ $= \frac{4x+6}{(x-2)(x+2)} = \frac{2(2x+3)}{x^2-4}$	2 Marks: Correct answer. 1 Mark: Finds a common denominator or recognises the difference of two squares.
12(b)	$\frac{1}{\sqrt{5}-1} = \frac{1}{\sqrt{5}-1} \times \frac{\sqrt{5}+1}{\sqrt{5}+1}$ $= \frac{\sqrt{5}+1}{5-1} = \frac{\sqrt{5}+1}{4}$	2 Marks: Correct answer. 1 Mark: Uses the conjugate.
12(c)	$(1-y)^3 + (1+y)^2 = (1-y)(1-2y+y^2) + (1+2y+y^2)$ $= 1-2y+y^2 - y+2y^2 - y^3 + 1+2y+y^2$ $= 2-y+4y^2-y^3$	2 Marks: Correct answer. 1 Mark: Expands one binomial.
12(d) (i)	In $\triangle ABC$ and $\triangle ADE$ $\angle CAB = \angle DAE$ (common angle) $\angle BCA = \angle EDA$ (corresponding angles are equal, parallel lines) $\angle CBA = \angle DEA$ (corresponding angles are equal, parallel lines) $\triangle ABC$ is similar to $\triangle ADE$ (equiangular)	2 Marks: Correct answer. 1 Mark: One or more correct statements.
12(d) (ii)	$\frac{AE}{AB} = \frac{AD}{AC} \quad (\text{matching sides in similar triangles})$ $\frac{AE}{8} = \frac{9}{3+9}$ $AE = \frac{9}{12} \times 8 = 6$ $BE = AB - AE = 8 - 6 = 2$	2 Marks: Correct answer. 1 Mark: Uses the corresponding sides in similar triangles.
12(e) (i)	$\angle OCD = 60^\circ$ (all angles are equal, equilateral triangle $\triangle ODC$) $\angle ACD = (60-x)^\circ$ (adjacent angles) $\angle ACB = 60^\circ$ (all angles are equal, equilateral triangle $\triangle ABC$) $\angle BCO = (60-x)^\circ$ (adjacent angles) $\therefore \angle ACD = \angle BCO$ (both equal to $(60-x)^\circ$)	1 Mark: Correct answer.
12(e) (ii)	In $\triangle ACD$ and $\triangle BCO$ $\angle ACD = \angle BCO$ (from part (i)) $AC = BC$ (all sides are equal, equilateral triangle $\triangle ABC$) $CD = CO$ (all sides are equal, equilateral triangle $\triangle ODC$) $\therefore \triangle ACD \equiv \triangle BCO$ (SAS)	2 Marks: Correct answer. 1 Mark: Correct argument insufficient justification.

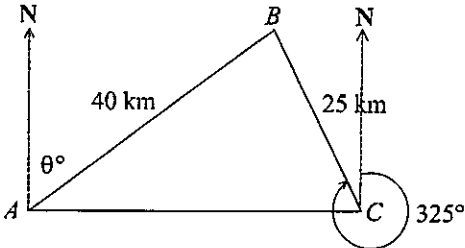
Question 13 (16 marks)

13(a) (i)	$x^2 - 16 = 0$ $(x + 4)(x - 4) = 0$ $\therefore x = \pm 4$	1 Mark: Correct answer.
13(a) (ii)	$25^k \cdot (5^3)^4 = 1$ $(5^2)^k \times (5^3)^4 = 5^0$ $2k + 12 = 0$ $2k = -12$ $k = -6$	2 Marks: Correct answer. 1 Mark: Makes some progress.
13(b) (i)	$y^3 + 10y^2 + 25y = y(y^2 + 10y + 25)$ $= y(y + 5)^2$	1 Mark: Correct answer.
13(b) (ii)	$a^3 - \frac{1}{8} = a^3 - \left(\frac{1}{2}\right)^3$ $= \left(a - \frac{1}{2}\right)\left(a^2 + \frac{1}{2}a + \frac{1}{4}\right)$	1 Mark: Correct answer.
13(c) (i)		1 Mark: Correct answer.
13(c) (ii)		1 Mark: Correct answer.
13(c) (iii)		1 Mark: Correct answer.

13(d)	$\lim_{x \rightarrow 4} \frac{x^2 - x - 12}{x - 4} = \lim_{x \rightarrow 4} \frac{(x - 4)(x + 3)}{(x - 4)}$ $= \lim_{x \rightarrow 4} (x + 3) = 7$	2 Marks: Correct answer. 1 Mark: Factorises the quadratic.
13(e)	$f(3) + f(0) + f(-3) = 3^3 + 1 - \frac{1}{3}$ $= 27\frac{2}{3}$	2 Marks: Correct answer. 1 Mark: Finds the value of one function.
13(f) (i)	$y^2 = x + 9 \quad (1)$ $x - 3y + 9 = 0 \quad (2)$ Make x the subject of equation (1) $x = y^2 - 9 \quad (3)$ Substitute $y^2 - 9$ for x into equation (2) $y^2 - 9 - 3y + 9 = 0$ $y(y - 3) = 0$ $\therefore y = 0 \text{ or } y = 3$ Substitute $y = 0$ into equation (3) $x = 0^2 - 9 = -9$ Substitute $y = 3$ into equation (3) $x = 3^2 - 9 = 0$ Points A and B are $(0, 3)$ and $(-9, 0)$.	2 Marks: Correct answer. 1 Mark: Finds one of the solutions or shows some understanding.
13(f) (ii)	$24 + 2x - x^2 \leq 0$ $(6 - x)(4 + x) \leq 0$  $\therefore x \leq -4 \text{ or } x \geq 6$	2 Marks: Correct answer. 1 Mark: Factorises the quadratic.

Question 14 (16 marks)

14(a)	$\begin{aligned} \text{LHS} &= \tan \theta - \tan \theta \sin^2 \theta \\ &= \tan \theta (1 - \sin^2 \theta) \\ &= \frac{\sin \theta}{\cos \theta} \times \cos^2 \theta \\ &= \sin \theta \cos \theta = \text{RHS} \end{aligned}$	2 Marks: Correct answer. 1 Mark: Uses relevant trig identity.
14(b)	$\begin{aligned} 2x + y - 1 &= 0 & (1) \\ 3x - y - 4 &= 0 & (2) \end{aligned}$ Equation (1) + (2) $5x - 5 = 0$ $x = 1$ Substitute $x = 1$ into equation (1) $2 \times 1 + y - 1 = 0 \text{ or } y = -1$ Solution is $x = 1$ and $y = -1$.	2 Marks: Correct answer. 1 Mark: Finds one of the solutions.
14(c) (i)	$\begin{aligned} \frac{d}{dx} \sqrt{x^4 + 7} &= \frac{d}{dx} (x^4 + 7)^{\frac{1}{2}} = \frac{1}{2} (x^4 + 7)^{-\frac{1}{2}} \times 4x^3 \\ &= \frac{2x^3}{\sqrt{x^4 + 7}} \end{aligned}$	1 Mark: Correct answer.
14(c) (ii)	$\begin{aligned} \frac{d}{dx} x(1-x)^6 &= x \times 6(1-x)^5 \times -1 + (1-x)^6 \times 1 \\ &= (1-x)^5 [-6x + 1(1-x)] \\ &= (1-x)^5 (1-7x) \end{aligned}$	2 Marks: Correct answer. 1 Mark: Uses the product rule with one correct value.
14(c) (iii)	$\begin{aligned} \frac{d}{dx} \left(\frac{3x}{2-x} \right) &= \frac{(2-x) \times 3 - 3x \times -1}{(2-x)^2} \\ &= \frac{6 - 3x + 3x}{(2-x)^2} = \frac{6}{(2-x)^2} \end{aligned}$	2 Marks: Correct answer. 1 Mark: Uses the quotient rule with one correct value.
14(d)	$y = x^2 - 6x, \quad \frac{dy}{dx} = 2x - 6$ At $(-1, 7) \quad \frac{dy}{dx} = 2 \times -1 - 6 = -8$ Equation of the tangent at $(-1, 7)$ with $m = -8$ $\begin{aligned} y - y_1 &= m(x - x_1) \\ y - 7 &= -8(x - (-1)) \\ y - 7 &= -8x - 8 \\ 8x + y + 1 &= 0 \end{aligned}$	2 Marks: Correct answer. 1 Mark: Finds the derivative or shows some understanding.

14(e)	$2^x \times 4^{x+1} = \frac{1}{2} \text{ or } 2^x \times 2^{2(x+1)} = 2^{-1}$ $2^{x+2(x+1)} = 2^{-1}$ $x + 2x + 2 = -1$ $3x = -3$ $x = -1$	1 Mark: Correct answer.
14(f) (i)	$\angle ACB = 325^\circ - 270^\circ = 55^\circ$ 	1 Mark: Correct answer.
14(f) (ii)	In $\triangle ABC$ use sine rule. $\frac{\sin \angle BAC}{25} = \frac{\sin 55^\circ}{40}$ $\sin \angle BAC = \frac{25 \times \sin 55^\circ}{40}$ $\angle BAC = 30.79514161 \approx 31^\circ$ Bearing $= 90^\circ - 31^\circ = 059^\circ$	3 Marks: Correct answer. 2 Marks: Finds $\angle BAC$. 1 Mark: Uses the sine rule with one correct value.